

Lecture 2: Cardinality

Mathematical Analysis

1 Intervals

Definition 1.1 (Interval). [def:interval] A set $I \subseteq \mathbb{R}$ is an *interval* if, whenever $x, y \in I$ with $x \leq y$, every $z \in \mathbb{R}$ satisfying $x \leq z \leq y$ also belongs to I .

Remark 1.1 (Intervals as Sets). [rem:intervals-as-sets] In these notes, every interval is a set contained in $\mathbb{R}$; that is, every interval I satisfies $I \subseteq \mathbb{R}$.

Definition 1.2 (Open Interval). [def:open-interval] For $a, b \in \mathbb{R}$ with $a < b$, the *open interval* from a to b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition 1.3 (Closed Interval). [def:closed-interval] For $a, b \in \mathbb{R}$ with $a \leq b$, the *closed interval* from a to b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Lemma 1.1 (Endpoints of Contained Closed Intervals). [lem:contained-interval-endpoints] Let $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ be closed intervals. If $I_1 \subseteq I_2$, then

$$a_2 \leq a_1 \leq b_1 \leq b_2.$$

Proof. Since $a_1, b_1 \in I_1 \subseteq I_2$, definition 1.3 gives $a_2 \leq a_1$ and $b_1 \leq b_2$.

Since $I_1 = [a_1, b_1]$ is a closed interval, definition 1.3 requires $a_1 \leq b_1$.

Combining these inequalities gives the result. □

Theorem 1.1 (Nested Intervals Property (NIP)). [thm:nip] Let $I_n = [a_n, b_n]$ be a sequence of closed and bounded intervals.

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \implies \bigcap_{n=0}^{\infty} I_n \neq \emptyset.$$

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$. The nesting hypothesis gives

$$\forall n \in \mathbb{N}, \quad I_{n+1} \subseteq I_n.$$

For fixed $p \in \mathbb{N}$, define

$$P_p(r) \iff I_{p+r} \subseteq I_p.$$

$$P_p(0), \quad P_p(r) \implies I_{p+r+1} \subseteq I_{p+r} \subseteq I_p \implies P_p(r+1).$$

By Natural Number Induction and $p \leq q \iff \exists r \in \mathbb{N} : q = p + r$,

$$\forall p, q \in \mathbb{N}, \quad p \leq q \implies \exists r \in \mathbb{N} : \quad q = p + r \wedge I_q = I_{p+r} \subseteq I_p.$$

By comparability in an ordered set, $\forall n, m \in \mathbb{N}$, $(n \leq m) \vee (m \leq n)$. Hence lemma 1.1 gives

$$n \leq m \implies I_m \subseteq I_n \implies a_n \leq a_m \leq b_m,$$

$$m \leq n \implies I_n \subseteq I_m \implies a_n \leq b_n \leq b_m,$$

so

$$\forall n, m \in \mathbb{N}, \quad a_n \leq b_m.$$

Consequently,

$$a_0 \in A, \forall n \in \mathbb{N}, a_n \leq b_0 \implies A \neq \emptyset \wedge A \text{ is bounded above.}$$

By the Axiom of Completeness, choose $\alpha = \sup A$. The definition of supremum gives

$$\forall n, m \in \mathbb{N}, \quad a_n \leq \alpha \leq b_m.$$

Hence, by the definitions of a closed interval and indexed intersection,

$$(\forall n \in \mathbb{N}, \quad a_n \leq \alpha \leq b_n) \implies (\forall n \in \mathbb{N}, \quad \alpha \in I_n) \iff \alpha \in \bigcap_{n=0}^{\infty} I_n.$$

Therefore $\bigcap_{n=0}^{\infty} I_n \neq \emptyset$. □

Remark 1.2 (Intervals can be Infinitely Small). [rem:inf-small-intervals] If a size of the interval is infinitely small, ie $I = [a, b]$, $\forall \varepsilon > 0$, $b - a < \varepsilon$, this means that the interval is equivalent to a single point, ie $a = b$.

2 Decimal Expansion

Theorem 2.1 (Decimal Expansion). [thm:decimal-representation] Let $D = \{0, 1, \dots, 9\}$. Every digit sequence $a : \mathbb{N}_1 \rightarrow D$ determines a unique $x \in [0, 1]$:

$$\forall a : \mathbb{N}_1 \rightarrow D, \quad \exists! x \in [0, 1], \quad x = \sum_{k=1}^{\infty} a_k 10^{-k}.$$

Proof. We call the summation of the first n decimals s_n , ie

$$s_n = \sum_{k=1}^n a_k 10^{-k}$$

Since $\forall k \in \mathbb{N}_1, 0 \leq a_k \leq 9$, meaning that

$$\forall n, m \in \mathbb{N}_1, n \leq m \implies s_m = s_n + \sum_{k=n+1}^m a_k 10^{-k} \leq s_n + 9 \sum_{k=n+1}^m 10^{-k} = s_n + 10^{-n} - 10^{-m} < s_n + 10^{-n}$$

In other words,

$$\forall n, m \in \mathbb{N}_1, \quad n \leq m \implies s_m \in [s_n, s_n + 10^{-n}]$$

By the same argument, we can also show that

$$\forall n, m \in \mathbb{N}_1, \quad n \leq m \implies s_m + 10^{-m} \in [s_n, s_n + 10^{-n}]$$

Where we can construct the following nested intervals

$$I_n = [s_n, s_n + 10^{-n}], \quad I_1 \supseteq I_2 \supseteq \dots, \quad x = \sum_{n=\mathbb{N}_1} a_n 10^{-k} = \bigcap_{n \in \mathbb{N}_1} I_n$$

According to the Nested Interval Theorem, after reindexing,

$$\bigcap_{n \in \mathbb{N}_1} I_n \neq \emptyset.$$

Also, notice that $I_n = [s_n, s_n + 10^{-n}]$, $s_n + 10^{-n} - s_n = 10^{-n}$. According to the Remark, we can conclude:

$$\forall \varepsilon > 0, \exists n \in \mathbb{N}_1 : I_n = [s_n, s_n + 10^{-n}] \text{ and } (s_n + 10^{-n}) - s_n = 10^{-n} < \varepsilon \implies \exists! x \in \bigcap_{n \in \mathbb{N}_1} I_n$$

Thus x is the unique number represented by the digit sequence a . □

Remark 2.1 (Decimal Expansion is not Unique). [rem:decimal-representation-not-unique] The decimal expansion of a real number is not unique. For example, $0.8999\dots = 0.9000\dots$

3 Archimedean Property

Theorem 3.1 (Archimedean Property). [thm:archimedean-property] $\mathbb{N}$ is not bounded above in $\mathbb{R}$, in other words, $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, n > x$

Proof. Assume for the sake of contradiction that $\mathbb{N}$ is bounded above in $\mathbb{R}$.

Since $0 \in \mathbb{N}$, we have $\mathbb{N} \neq \emptyset$. By the Axiom of Completeness, $\mathbb{N}$ has a supremum, denoted as $\alpha = \sup \mathbb{N}$.

By the definition of epsilon criteria, we have:

$$\forall x \in \mathbb{N}, x \leq \alpha, \quad \forall \varepsilon > 0, \exists x \in \mathbb{N}, \alpha - \varepsilon < x$$

Taking $\varepsilon = 1$, we obtain

$$\exists n \in \mathbb{N} : \quad \alpha - 1 < n \implies \alpha < n + 1.$$

Since $n + 1 = S(n) \in \mathbb{N}$, this contradicts that α is an upper bound of $\mathbb{N}$, namely

$$\forall m \in \mathbb{N}, \quad m \leq \alpha.$$

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 3.2 (Archimedean Property (Reciprocal Form)). [thm:archimedean-reciprocal-form]

$$\forall y > 0, \exists n \in \mathbb{N}, 1/n < y$$

Proof. Take $x = 1/y$. Since $y > 0$, we have $x > 0$.

According to definition of Natural Numbers, $\exists n \in \mathbb{N}, n \geq x$, then $1/n \leq 1/x = y$. □

Definition 3.1 (Dense Subset). [def:dense-subset] Let $D \subseteq \mathbb{R}$. We say that D is *dense in* $\mathbb{R}$ if

$$\forall a, b \in \mathbb{R}, \quad a < b \implies \exists d \in D : \quad a < d < b.$$

Theorem 3.3 (Density of the Rational Numbers). [thm:rational-dense-in-reals] $\mathbb{Q}$ is dense in $\mathbb{R}$.

Theorem 3.4 (Density of the Irrational Numbers). [thm:irrational-dense-in-reals] $\mathbb{R} \setminus \mathbb{Q}$ (ie, $\mathbb{I}$) is dense in $\mathbb{R}$.

4 Counting